

NE 470/570 Monte Carlo Methods for Radiation Transport

Topic 1: Random Variables and Sampling

Xu Wu

xwu27@ncsu.edu

Associate Professor
Department of Nuclear Engineering
North Carolina State University

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Outline

1. Random Variables and Random Numbers
2. Fundamental Formulation of Monte Carlo
3. Sampling from 1-D PDF
4. Sampling from n-D PDF
5. Sampling Examples

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1. Random Variables and Random Numbers
2. Fundamental Formulation of Monte Carlo
3. Sampling from 1-D PDF
4. Sampling from n-D PDF
5. Sampling Examples

Outline

1. Random Variables and Random Numbers

1.1. Probability Space

1.2. Random Variables

1.3. Random Numbers

Understand the **probability space** or a **probability triple** $(\Omega, \mathcal{F}, \mathcal{P})$

- In probability theory, a probability space or a probability triple $(\Omega, \mathcal{F}, \mathcal{P})$ is a mathematical construct that provides a formal model of a random process or “experiment”, where Ω is a sample space, $\mathcal{F}$ is a σ -algebra, and $\mathcal{P}$ is a probability measure.
- The set, Ω , of all possible outcomes of a particular experiment is called the **sample space**.
 - 1 If the experiment consists of **tossing a coin**, then the sample space is

$$\Omega = \{H, T\}$$

- 2 If the experiment consists of **throwing a die**, then the sample space is

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

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- 3 What is the sample space for the sum of throwing two dies together?
- 4 What is the sample space for neutron interaction with matter?

- A collection of all the subsets of Ω is the **event space**, also called a **sigma algebra** (or **σ -algebra**), or **Borel field**, denoted by $\mathcal{F}$.
- It has the following properties:
 - 1 **$\mathcal{F}$ contains the sample space**: $\Omega \in \mathcal{F}$;
 - 2 **$\mathcal{F}$ is closed under complements**: if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$, where $A^c = \Omega \setminus A$;
 - 3 **$\mathcal{F}$ is closed under countable unions**: if $A_i \in \mathcal{F}$ for $i = 1, 2, \dots$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.
- More specifically, if Ω is finite or countable, then $\mathcal{F} = \{\text{all subsets of } \Omega, \text{ including } \Omega \text{ itself}\}$.
 - For example, for throwing a coin, $\mathcal{F} = \{\{\emptyset\}, \{H\}, \{T\}, \{H, T\}\}$.
- If there are n events in Ω , then $\mathcal{F}$ contains **2^n elements**. (Why?)

- Given a sample space Ω and an associated σ -algebra $\mathcal{F}$, a **probability function/measure** is a function $\mathcal{P}$ which assigns each event in the event space a probability.
- $\mathcal{F}$ satisfies:
 - 1 $\mathcal{P}(A) \geq 0$ for all $A \in \mathcal{F}$;
 - 2 $\mathcal{P}(\Omega) = 1$;
 - 3 $\mathcal{P}$ is **countably additive** (also called **σ -additive**): if $A_1, A_2, \dots \in \mathcal{F}$ are **pairwise disjoint**, then

$$\mathcal{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathcal{P}(A_i) \quad (1)$$

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What is a Random Variable (RV)?

- In probability and statistics, a **random variable**, also called **random quantity**, **aleatory variable**, or **stochastic variable** is described informally as a variable taking on **numerical values** determined by the outcome of a random phenomenon.
- Why do we need numerical values for a RV?
 - Numerical values are easier to work with.
 - For example, you can easily count the total number of heads/tails from 10,000 trials of coin toss by assigning head=0 and tail=1.

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- Why do we need numerical values for a RV?
 - Numerical values are easier to work with.
 - For example, you can easily count the total number of heads/tails from 10,000 trials of coin toss by assigning head=0 and tail=1.
- In probability theory, a RV is formally defined as a **measurable function** on a probability space $(\Omega, \mathcal{F}, \mathcal{P})$ that maps from the sample space Ω to the **real numbers**.
 - The **sample space** is also called the **domain** of the RV, defined as the set of possible outcomes of a random event.
 - For example, in the event of a coin toss, only two outcomes are possible: heads or tails. A RV can map these outcomes to 0 and 1, respectively.

What is a Random Variable (RV)?

- A RV X is a measurable function $X : \Omega \rightarrow E$ from a set of possible outcomes in Ω to a measurable space E .
- The probability that X takes on a value in a measurable set $S \subseteq E$ is written as:

$$P(X \in S) = P(\{\omega \in \Omega | X(\omega) \in S\}) \quad (2)$$

where X is usually real-valued numbers, so $E = \mathbb{R}$, the **real space**.

- In this course, we **use capital letters like X to represent a RV**, and **lower case letters x to represent a realization of the RV**.

- A RV has a **probability distribution**, which is the mathematical function that gives the probabilities of occurrence of different possible outcomes for an experiment.
- In other words, the probability distribution of a RV X tells us what the possible values of X are and what probabilities are assigned to those values.

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- When the possible values of X is **countable**, the RV is called a **discrete RV**, which can be described by a **probability mass function (PMF)**.
- When the possible values of X is **uncountable**, the RV is called a **continuous RV**, which can be described by a **probability density function (PDF)**.

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- When the possible values of X is **uncountable**, the RV is called a **continuous RV**, which can be described by a **probability density function (PDF)**.
- RVs are fundamental elements in a random physical process, which can generally be characterized by their **average behavior and associated statistical uncertainties**.
- **Monte Carlo** is a powerful tool to do this.

Discrete RV

- The **probability mass function (PMF)** $p(x)$ is the probability that a discrete RV X takes a value of x .
- For example, for a well-balanced cubical die, the probability of any event x_n , is:

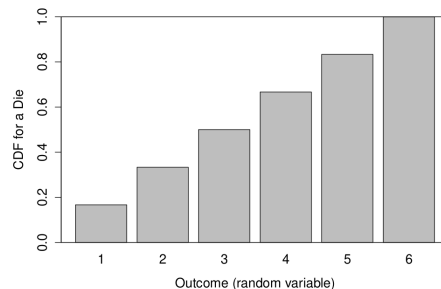
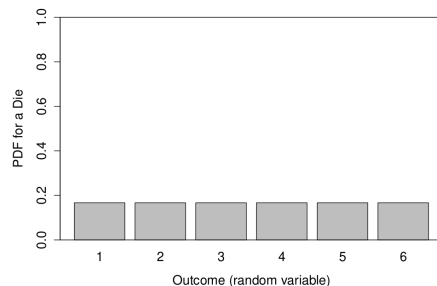
$$p(x_n) = \frac{1}{6}, \quad \text{for } n = 1, 2, \dots, 6$$

- Note that the PMF is **normalized** such that

$$\sum p(x_n)_{n=1}^N = 1$$

- The **cumulative distribution function (CDF)** $P(x)$ is the probability that X takes a value **not exceeding** x .

$$P(x_n) = \sum_{i=1}^n p(x_i) \quad (3)$$



Continuous RV

- For a continuous RV X that is defined on a range $[a, b]$, the **probability density function (PDF)** $p(x)$ is the probability that X will assume a value in the range of $[x, x + dx]$.
- The PDF $p(x)$ is also normalized.

$$\int_a^b p(x) dx = 1 \quad (4)$$

- The CDF $P(x)$ represents the probability that X will assume a value not exceeding x .

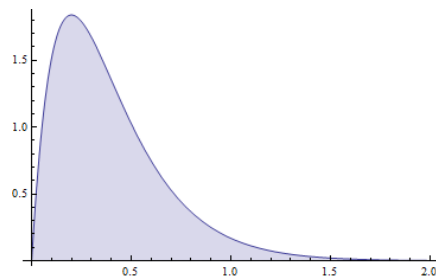
$$P(x) = \int_a^x p(x') dx' \quad (5)$$

Characteristics of PDF/PMF and CDF

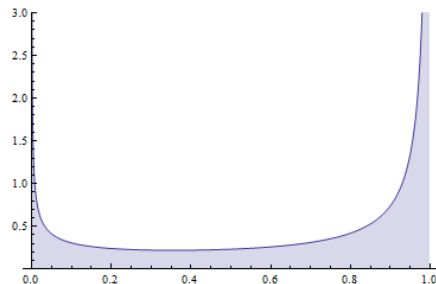
- 1 PDF/PMF is always non-negative.
- 2 CDF is always a positive and non-decreasing function of X .
- 3 PDF/PMF is normalized such that its corresponding CDF varies in a range of $[0, 1]$.
 - Note that this does not mean PMF/PDF values are always between $[0, 1]$. The values of PMF will indeed always be in $[0, 1]$. However, the values of PDF can be any non-negative number, even infinite.

Example PDFs

- **Upper:** this example shows the PDF for a **Gamma distribution** (with shape parameter of $3/2$ and scale of $1/5$). Because most of the x is less than 1, the curve has to rise higher than 1 in order to have a total area of 1 as required for the normalization property. That is, the total shaded area should be 1.

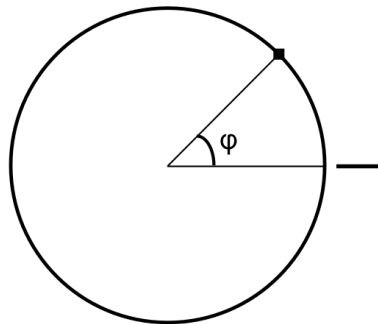


- **Lower:** this example shows a PDF for a **beta distribution** with parameters $1/2, 1/10$ becomes infinite at 0 and at 1. The total area still is finite (and equals 1).



Continuous RV example

- Consider a marked disk. Every time it spins, the angle ϕ between the **mark** and a **reference position** can be considered as a continuous RV in the range of $[0, 2\pi]$.
- For a well balanced disk, the probability of getting any ϕ should be independent of ϕ , i.e., constant, $p(\phi) = k$.



Continuous RV example

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- For a well balanced disk, the probability of getting any ϕ should be independent of ϕ , i.e., constant, $p(\phi) = k$.
- Due to the normalization constraint:

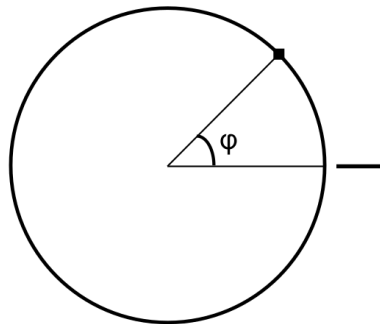
$$\int_0^{2\pi} p(\phi) d\phi = 1 \rightarrow 2\pi k = 1, k = \frac{1}{2\pi} \quad (6)$$

- The PDF is:

$$p(\phi) = \frac{1}{2\pi}, \quad \phi \in [0, 2\pi] \quad (7)$$

- The CDF is:

$$P(\phi) = \int_0^{\phi} p(\phi') d\phi' = \frac{\phi}{2\pi}, \quad \phi \in [0, 2\pi] \quad (8)$$



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Random Numbers (RNs)

- **Random Numbers** are a sequence of numbers $\{\eta_1, \dots, \eta_n, \dots\}$ which have the special characteristic that it is not possible to predict η_{n+1} based on the previous n numbers $\{\eta_i\}_{i=1}^n$ in the sequence.
- To assure this **unpredictability (randomness)**, the numbers should pass **randomness tests**.
- To generate a sequence of RNs, we need a **generating approach/function** that yields numbers that are **uniformly** distributed in a range (**commonly** $[0, 1]$).
- In other words, a sequence of RNs follow a **standard uniform distribution**. (What are its PDF and CDF?)
- If $Y \sim \text{Uniform}[a, b]$, how to transform Y to a standard uniform RV?

Use the spinning disk to generate RNs

- ϕ follows a uniform distribution in the range $[0, 2\pi]$. Set

$$\eta = \frac{\phi - 0}{2\pi - 0} = \frac{\phi}{2\pi} \quad (9)$$

- Assume the PDF of η is $q(\eta)$. Because

$$|q(\eta)d\eta| = |p(\phi)d\phi| \quad (10)$$

we have

$$q(\eta) = p(\phi) \left| \frac{d\phi}{d\eta} \right| = \frac{1}{2\pi} \cdot 2\pi = 1 \quad (11)$$

- η follows a **uniform distribution in the range of $[0, 1]$** . Therefore, it is a valid RN generator.

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Fundamental Formulation of Monte Carlo (FFMC) - continuous RV

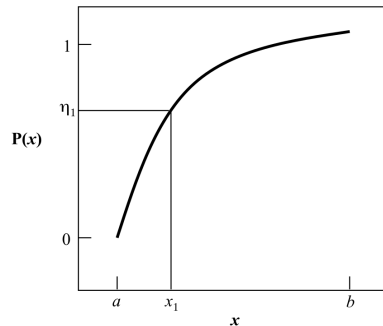
- The **goal of Monte Carlo simulation** is to simulate a physical process in which the basic physics is known, i.e., **we know the PDFs of the basic processes**.
- Assuming that we can generate the RNs, we want to obtain samples of X , i.e., sample the outcome based on $p(x)$.
- Assume that the RV X is related to the RN η

$$p(x)dx = q(\eta)d\eta, \quad \text{for } a \leq x \leq b \text{ and } 0 \leq \eta \leq 1 \quad (12)$$

- Integrate on both sides:

$$\int_a^x p(x')dx' = \int_0^\eta 1 \cdot d\eta' \quad \rightarrow \quad P(x) = \eta \quad (13)$$

- The above equation is called the **FFMC**, which gives a **relation for obtaining a continuous RV X using a RN η** .



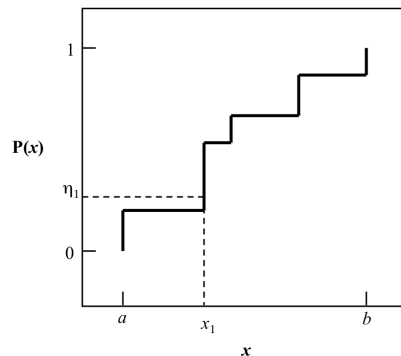
FFMC - discrete RV

- How do we deal with discrete RVs?
- Because the RN η is continuous, while a discrete PMF only assumes certain values, we have to set the following relation:

$$\eta = \min_{x_n} P(x_n) \geq \eta \quad (14)$$

where $P(x_n) = \sum_{i=1}^n p(x_i)$. This means that n is selected when the minimum of $P(x_n)$ is greater than or equal to η .

- Given the FFMC, we can solve it to generate samples for the corresponding discrete RV X .
- In next section, we will look at a few approaches for solving the FFMC for processes with a single RV, i.e., with one-dimensional density functions.



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3. Sampling from 1-D PDF

3.1. Analytical inversion

3.2. Numerical inversion

3.3. Probability mixing method

3.4. Rejection technique

3.5. Numerical evaluation

3.6. Table lookup

Analytical inversion

- The FFMC is **inverted** to obtain a formulation for a RV X in terms of a RN $\eta \in [0, 1]$.
- The **inverse formulation** is given by:

$$x = P^{-1}(\eta) \quad (15)$$

- However, P^{-1} is **not always available**. As the PDF becomes complex, obtaining $P^{-1}(\eta)$ analytically becomes more complicated or even impossible.

Simple example using an uniform distribution

- The PDF for a uniform RV X is given by $p(x) = \frac{1}{2}$ for $-1 \leq x \leq 1$, then the corresponding FPMC is given by:

$$P(x) = \int_{-1}^x p(x') dx' = \int_{-1}^x \frac{1}{2} dx' = \frac{1}{2}(x + 1) = \eta \quad (16)$$

- As a result of analytical inversion, x or $P^{-1}(\eta)$ is:

$$x = P^{-1}(\eta) = 2\eta - 1 \quad (17)$$

Analytical inversion is essentially the same with the **inverse transform sampling**

- **Inverse transform sampling** (also known as **inversion sampling**, the **inverse probability integral transform**, or the **inverse transformation method**) is a basic method for pseudo-random number sampling, i.e., for generating samples from **any** probability distribution given its CDF.
- Given a continuous RV X , with CDF $F_X(x)$. By definition:

$$F_X(x) = P(X \leq x) \quad (18)$$

- Another RV U is defined as $U = F_X(X)$. Then U follows a uniform distribution on $[0, 1]$, regardless of the distribution type of X . (What is the difference between $F_X(X)$ and $F_X(x)$?)

- This is because

$$F_U(u) = P(U \leq u) = P(F_X(X) \leq u) = P(X \leq F_X^{-1}(u)) = F_X(F_X^{-1}(u)) = u \quad (19)$$

which is exactly the CDF of a standard uniform distribution, since $U \in [0, 1]$.

- It is extremely easy to generate samples from a uniform distribution. Once we have uniform samples $\{u_i\}_{i=1}^N$, we can obtain samples for X using

$$x_i = F_X^{-1}(u_i) \quad (20)$$

as long as F_X is invertible.

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Numerical inversion

- In a numerical procedure, the PDF is partitioned into N equi-probable areas within $[a, b]$.
- In each area the probability is equal to:

$$\int_{x_{i-1}}^{x_i} p(x') dx' = \frac{1}{N} \quad (21)$$

- The average PDF within each interval is given by

$$p_i = \frac{1}{N(x_i - x_{i-1})}, \quad \text{for } i = 1, \dots, N \quad (22)$$

which is obtained by assuming a uniform distribution over each partition.

- Once the partition is determined, use the steps on the next page to sample X .

1 Generate two RNs (η_1 and η_2).

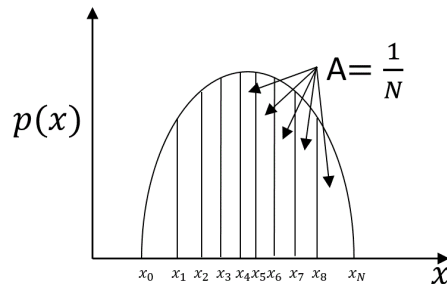
2 Sample an interval using:

$$i = \text{INT}(N \cdot \eta_1) + 1$$

where $\text{INT}()$ returns the integer part.

3 Sample X within the i^{th} interval using:

$$x = x_{i-1} + \eta_2 \cdot (x_i - x_{i-1})$$



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Probability mixing method

- If the PDF $p(x)$ can be partitioned into n **non-negative functions**:

$$p(x) = \sum_{i=1}^n f_i(x), \quad \text{for } f_i(x) \geq 0, \quad a \leq x \leq b \quad (25)$$

then, we may define PDFs corresponding to each $f_i(x)$ as

$$p_i(x) = \alpha_i f_i(x) \quad (26)$$

where α_i is a constant that normalizes $f_i(x)$.

- We will have the following relations. (Why?)

$$p(x) = \sum_{i=1}^n \frac{1}{\alpha_i} p_i(x), \quad \sum_{i=1}^n \frac{1}{\alpha_i} = 1 \quad (27)$$

- The **probability mixing method** is a two-step procedure for sampling the RV X :

1 Generate a **first RN** η_1 , and select the i^{th} PDF $p_i(x)$ using the following inequality:

$$\sum_{k=1}^{i-1} \frac{1}{\alpha_k} \leq \eta_1 < \sum_{k=1}^i \frac{1}{\alpha_k} \quad (28)$$

2 Generate a **second RN** η_2 , and then sample X from the selected i^{th} PDF $p_i(x)$, by:

$$\eta_2 = P_i(x) = \alpha_i \int_a^x f_i(x') dx' \quad (29)$$

- Note that **this method is only useful if the equation in Step 2 is easily solvable**. That is, each individual $f_i(x)$ should be analytically invertible.

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Rejection technique

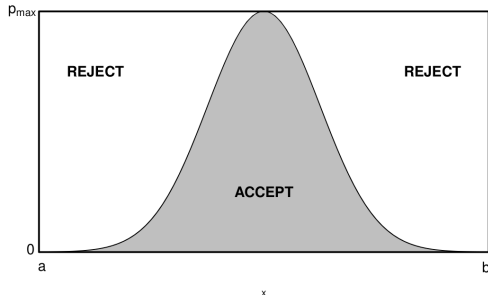
- When $P^{-1}(\eta)$ is **not available analytically**, a method called the **rejection technique** can be used:

- 1 Enclose $p(x)$ in a frame bounded by $p_{\max}$, a and b .
- 2 Generate two RNs η_1 and η_2 .
- 3 Sample the RV X using:

$$x = a + \eta_1 \cdot (b - a) \quad (30)$$

- 4 Accept x if

$$\eta_2 \cdot p_{\max} \leq p(x) \quad (31)$$



- Effectively, we are sampling from the area under the PDF, i.e., the CDF.
- Because the technique samples from the area, we can define an **efficiency formulation** as:

$$\text{efficiency} = \frac{\int_a^b p(x)dx}{p_{\max} \cdot (b - a)} = \frac{1}{p_{\max} \cdot (b - a)} \quad (32)$$

- Arbitrary $p_{\max}$ can be used, as long as it is above the top of $p(x)$.
- However, larger $p_{\max}$ will cause lower efficiency because more samples are rejected.

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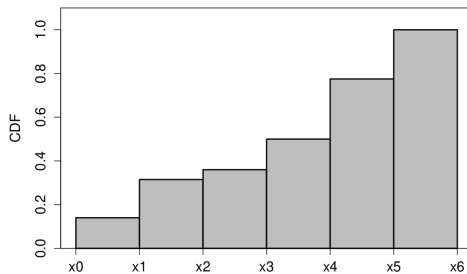
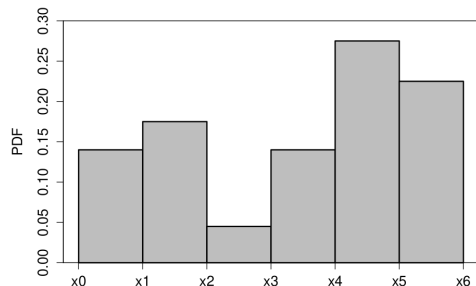
- If a continuous PDF is represented as a histogram, then its CDF is obtained using:

$$P_i = \sum_{k=1}^i p_k (x_k - x_{k-1}), \quad \text{for } i = 1, \dots, n \quad (33)$$

- To obtain a FFMC for a continuous RV, we develop an interpolation formulation given by:

$$P(x) = \frac{x - x_{i-1}}{x_i - x_{i-1}} P_i + \frac{x_i - x}{x_i - x_{i-1}} P_{i-1} \quad (34)$$

Can you draw this linear curve on the CDF figure?



- Then the FFMC for X is:

$$\eta = P(x) = \frac{x - x_{i-1}}{x_i - x_{i-1}} P_i + \frac{x_i - x}{x_i - x_{i-1}} P_{i-1} \quad (35)$$

- With an analytical inversion, X can be sampled as:

$$x = \frac{(x_i - x_{i-1})\eta - x_i P_{i-1} + x_{i-1} P_i}{P_i - P_{i-1}} \quad (36)$$

- To implement this method, the following procedure is used:

- 1 Generate a RN η .
- 2 Search for i such that $P_{i-1} < \eta < P_i$.
- 3 Generate a sample using the equation above.

- The second step in this procedure can use either **linear** or **binary** algorithms.

- In the **linear** search:
 - The search is performed by marching monotonically from the minimum value to the maximum value of the CDF until the necessary condition of the random variable is satisfied.
- In the **binary** search, the following algorithm is used:
 - 1 Generate a RN η
 - 2 The value of the CDF of the mid-number from the list of CDFs is identified.
 - 3 A half sequence is selected: (1) Lower-half sequence for $\eta > \text{CDF}(\text{mid-number})$, (2) Upper-half sequence for $\eta < \text{CDF}(\text{mid-number})$
 - 4 Iterate on steps 2 and 3, until the appropriate inequality for the RV is satisfied.
- It is important to note that computation times for the linear and binary searches are $\mathcal{O}(N)$ and $\mathcal{O}(\log 2(N))$, respectively. Hence, for large values of N , the binary search can be significantly more efficient.

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Table lookup

- In this approach, a table of CDF versus RV X is created and stored in the computer memory. Then sampling is performed by generating a RN η , which is compared with the CDF to determine the RV X .
- In the case of a continuous RV, the table entries resemble a histogram; therefore, the procedure discussed in “Numerical evaluation” should be applied.
- In the case of a discrete RV, the following procedure is used:
 - 1 A RN η is generated.
 - 2 A search for i is conducted such that the $P_{i-1} < \eta < P_i$ inequality is satisfied. Note that in this step, a **linear** or **binary** search algorithm can be employed as discussed in the “Numerical evaluation” method.
 - 3 Since it is a discrete RV, x_i is immediately available once i is identified. We won't use the formula in “Numerical evaluation” for continuous RV.

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4. Sampling from n-D PDF

4.1. Methodology

4.2. Example

Sampling from multi-dimensional PDFs

- We have introduced six techniques for solving the FPMC formulation corresponding to a **one-dimensional PDF** (with a single RV). Now we will develop FPMC formulations for sampling multiple RVs from **multi-dimensional PDFs**.
- Consider a general multi-dimensional PDF:

$$f(x_1, x_2, \dots, x_n) \quad (37)$$

where $x_1 \in [a_1, b_1]$, $x_2 \in [a_2, b_2]$, $\dots$, $x_n \in [a_n, b_n]$

- We need to derive one-dimensional PDF for each dimension in order to sample each RV.
- Starting with the **first RV X_1** , the PDF is given by:

$$\begin{aligned} p_1(x_1) &= \frac{\int_{a_2}^{b_2} dy_2 \int_{a_3}^{b_3} dy_3 \dots \int_{a_n}^{b_n} dy_n f(\mathbf{x}_1, y_2, \dots, y_n)}{\int_{a_1}^{b_1} dy_1 \int_{a_2}^{b_2} dy_2 \int_{a_3}^{b_3} dy_3 \dots \int_{a_n}^{b_n} dy_n f(y_1, y_2, \dots, y_n)} \\ &= \int_{a_2}^{b_2} dy_2 \int_{a_3}^{b_3} dy_3 \dots \int_{a_n}^{b_n} dy_n f(\mathbf{x}_1, y_2, \dots, y_n) \end{aligned} \quad (38)$$

because the integration in the denominator equals 1.

- For the **second RV** X_2 , the conditional PDF given X_1 , is defined as:

$$\begin{aligned}
 p_2(x_2|x_1) &= \frac{\int_{a_3}^{b_3} dy_3 \int_{a_4}^{b_4} dy_4 \dots \int_{a_n}^{b_n} dy_n f(\mathbf{x}_1, \mathbf{x}_2, y_3, \dots, y_n)}{\int_{a_2}^{b_2} dy_2 \int_{a_3}^{b_3} dy_3 \int_{a_4}^{b_4} dy_4 \dots \int_{a_n}^{b_n} dy_n f(\mathbf{x}_1, y_2, y_3, \dots, y_n)} \\
 &= \frac{\int_{a_3}^{b_3} dy_3 \int_{a_4}^{b_4} dy_4 \dots \int_{a_n}^{b_n} dy_n f(\mathbf{x}_1, \mathbf{x}_2, y_3, \dots, y_n)}{p_1(x_1)}
 \end{aligned} \tag{39}$$

- For the **third RV** X_3 , the conditional PDF given X_1 and X_2 , is defined as:

$$\begin{aligned}
 p_3(x_3|x_1, x_2) &= \frac{\int_{a_4}^{b_4} dy_4 \int_{a_5}^{b_5} dy_5 \dots \int_{a_n}^{b_n} dy_n f(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, y_4, \dots, y_n)}{\int_{a_3}^{b_3} dy_3 \int_{a_4}^{b_4} dy_4 \int_{a_5}^{b_5} dy_5 \dots \int_{a_n}^{b_n} dy_n f(\mathbf{x}_1, \mathbf{x}_2, y_3, \dots, y_n)} \\
 &= \frac{\int_{a_4}^{b_4} dy_4 \int_{a_5}^{b_5} dy_5 \dots \int_{a_n}^{b_n} dy_n f(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, y_4, \dots, y_n)}{p_2(x_2|x_1) \cdot p_1(x_1)}
 \end{aligned} \tag{40}$$

- Similarly, for the n^{th} RV X_n , the conditional PDF given $X_1, X_2, \dots, X_{n-1}$, is defined as:

$$p_n(x_n|x_1, x_2, \dots, x_{n-1}) = \frac{f(x_1, x_2, \dots, x_n)}{\int_{a_n}^{b_n} dy_n f(x_1, x_2, x_3 \dots, x_{n-1}, y_n)} \quad (41)$$

- The corresponding FFMCs for a multidimensional PDF are given by:

$$\begin{aligned} P_1(x_1) &= \int_{a_1}^{x_1} p_1(y_1) dy_1 = \eta_1 \\ P_2(x_2) &= \int_{a_2}^{x_2} p_2(y_2|x_1) dy_2 = \eta_2 \\ P_3(x_3) &= \int_{a_3}^{x_3} p_3(y_3|x_1, x_2) dy_3 = \eta_3 \\ &\vdots \\ P_n(x_n) &= \int_{a_n}^{x_n} p_n(y_n|x_1, \dots, x_{n-1}) dy_n = \eta_n \end{aligned} \quad (42)$$

- For each dimension, the one-dimensional FFMC solution methods can be used.

Outline

4. Sampling from n-D PDF

4.1. Methodology

4.2. Example

Example for sampling from multi-dimensional density function

- Selection of polar and azimuthal angles given the two-dimensional PDF:

$$p(\mu, \phi) = \frac{1}{4\pi}, \quad -1 \leq \mu \leq 1, 0 \leq \phi \leq 2\pi \quad (43)$$

Example for sampling from multi-dimensional density function

- Considering $X_1 = \mu$, $X_2 = \phi$:

$$p_1(\mu) = \frac{\int_0^{2\pi} d\phi p(\mu, \phi)}{\int_{-1}^1 d\mu \int_0^{2\pi} d\phi p(\mu, \phi)} = \int_0^{2\pi} d\phi \frac{1}{4\pi} = \frac{1}{2}, \quad -1 \leq \mu \leq 1 \quad (44)$$

$$p_2(\phi|\mu) = \frac{p(\mu, \phi)}{\int_0^{2\pi} d\phi p(\mu, \phi)} = \frac{\frac{1}{4\pi}}{\frac{1}{2}} = \frac{1}{2\pi}, \quad 0 \leq \phi \leq 2\pi \quad (45)$$

- Therefore, the corresponding FFMCS are given by:

$$P_1(\mu) = \int_{-1}^{\mu} p_1(y_1) dy_1 = \int_{-1}^{\mu} \frac{1}{2} dy_1 = \frac{\mu + 1}{2} = \eta_1, \quad \mu = 2\eta_1 - 1 \quad (46)$$

$$P_2(\phi) = \int_0^{\phi} p_2(y_2|\mu) dy_2 = \int_0^{\phi} \frac{1}{2\pi} dy_2 = \frac{\phi}{2\pi} = \eta_2, \quad \phi = 2\pi\eta_2 \quad (47)$$

Outline

1. Random Variables and Random Numbers
2. Fundamental Formulation of Monte Carlo
3. Sampling from 1-D PDF
4. Sampling from n-D PDF
5. Sampling Examples

Outline

5. Sampling Examples

5.1. Box-Muller procedure for Normal distribution

Box-Muller procedure for Normal distribution

- The normal/Gaussian distribution is the most commonly used distribution in modeling many physical phenomena. The PDF of a standard normal distribution is defined as:

$$p(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}, \quad -\infty < t < \infty \quad (48)$$

- Consider two **independent** RVs X and Y that follow normal distributions. The joint PDF of these two RVs is:

$$p(x, y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} = \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}}, \quad -\infty < x < \infty, -\infty < y < \infty \quad (49)$$

- We can express (x, y) in terms of **polar coordinates** (r, θ) , where r and θ are the **radial and angular coordinates**, respectively:

$$x = r \cdot \cos \theta$$

$$y = r \cdot \sin \theta$$

where $0 \leq r < \infty, 0 \leq \theta \leq 2\pi$.

- The **differential area becomes** $dxdy = r dr d\theta$. The joint PDF can be expressed in the polar coordinates as:

$$p(r, \theta) dr d\theta = p(x, y) dx dy = \frac{1}{2\pi} e^{-\frac{r^2}{2}} r dr d\theta = \left(e^{-\frac{r^2}{2}} r dr \right) \cdot \left(\frac{1}{2\pi} d\theta \right) \quad (50)$$

- Because $p(r, \theta)$ can be factored into the multiplication of two PDFs, r and θ are **independent**.

- We can also use the **theorem of probability functional transform**. Suppose (r, θ) has a joint PDF of $p(r, \theta)$, and (x, y) are functions of (r, θ) . Then $p(x, y)$ is defined as:

$$p(x, y) = \frac{p(r, \theta)}{\begin{vmatrix} \partial x / \partial r & \partial x / \partial \theta \\ \partial y / \partial r & \partial y / \partial \theta \end{vmatrix}} \quad (51)$$

- Given $\frac{\partial x}{\partial r} = \cos \theta$, $\frac{\partial x}{\partial \theta} = -r \sin \theta$, $\frac{\partial y}{\partial r} = \sin \theta$, $\frac{\partial y}{\partial \theta} = r \cos \theta$, we have:

$$p(x, y) = \frac{p(r, \theta)}{\begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}} = \frac{p(r, \theta)}{r} \quad (52)$$

- We will again have

$$p(r, \theta) = \frac{r}{2\pi} e^{-\frac{x^2+y^2}{2}} = \frac{r}{2\pi} e^{-\frac{r^2}{2}}, \quad 0 \leq r < \infty, 0 \leq \theta \leq 2\pi \quad (53)$$

- The procedure below shows the formulas for sampling from a two-dimensional normal distribution. (Pay attention to the sampling formula of r)

Table 2.2 Procedure for sampling from a normal distribution

$$p(\theta) = \frac{1}{2\pi} \quad \int_0^\theta d\theta \frac{1}{2\pi} = \eta_1 \quad \text{Sample } \theta = 2\pi\eta_1$$

$$p(r) = re^{-\frac{r^2}{2}} \quad \int_0^r dr r e^{-\frac{r^2}{2}} = \eta_2 \quad \text{Sample } r = \sqrt{-2 \ln \eta_2}$$

$$\text{Sample } x, y \quad x = \sqrt{-2 \ln \eta_2} \cos(2\pi\eta_1) \quad y = \sqrt{-2 \ln \eta_2} \sin(2\pi\eta_1)$$
